Isratul Ibnat Orpa

Dhaka, Bangladesh
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Professional Summary

Computer Science and Engineering undergraduate at AIUB with a strong foundation in mathematics and analytical problem-solving. Interested in software development, artificial intelligence, machine learning, and research. Co-author of two machine learning papers on explainable intrusion detection and phishing URL detection, presented at ICACT, Nepal. Committed to solving real-world problems through technology with a research-oriented and collaborative mindset.

Education

American International University-Bangladesh (AIUB) **2023 – Present**
Bachelor of Science in Computer Science and Engineering *Dhaka, Bangladesh*

Relevant Coursework

- Advanced Programming in Python
- Object-Oriented Programming
- Data Structures & Algorithms
- Software Engineering
- Database Systems
- Operating Systems
- Web Technologies
- Artificial Intelligence

Technical Projects

Personal Expense Tracker | *Python (Flask), SQLite, HTML5, CSS3, JavaScript* GitHub **October 2026**

- Built a minimal full-stack expense tracker with a Flask REST backend, SQLite persistence, and a Vanilla JS frontend.
- Implemented income/expense CRUD operations, category-based filtering, and a dashboard summarising monthly spending.

Online Medicine Shop | *PHP, MySQL, JavaScript (AJAX), HTML5, CSS3* GitHub **Spring 2026**

- Built Task 3 (cart, checkout, payment, and bill) for a team pharmacy web shop using PHP and MySQL.
- Implemented AJAX-powered product search; project follows parameterized SQL and salted password hashing throughout.

Intrusion Detection System | *Python, scikit-learn, pandas, SHAP, SMOTE* GitHub **2025**

- Implemented machine learning models for EV charging-station authentication attack detection as part of published research.
- Applied leakage-aware SMOTE, removed 10 leakage-prone features, and used SHAP for model explainability (Macro F1 0.9748).

Research & Publications

Machine Learning Research **Since 2025**
Undergraduate Researcher, Department of Computer Science, AIUB *Dhaka, Bangladesh*

- **Intrusion Detection:** applied leakage-free SMOTE and four ML models for EV charging-station authentication attacks.
- Submitted three research paper abstracts; two accepted and presented at ICACT, Nepal (proceedings forthcoming).

Publications

1. N. I. Rishad, F. Mahmud, **I. I. Orpa**, R. A. M. Rudro, and K. Nur, "Explainable and Leakage-Free Intrusion Detection for EV Charging Station Authentication Attacks," *ICACT, Nepal*

Technical Skills

Languages: Python, PHP, JavaScript, C++, C#, SQL, HTML5, CSS3
Frontend: HTML5, CSS3, JavaScript, AJAX
Backend & Databases: Flask, PHP, MySQL, SQLite, REST API design, parameterized queries
Tools & Platforms: Git, GitHub, VS Code, Cisco Packet Tracer, XAMPP
Data Science & ML: scikit-learn, pandas, Random Forest, SHAP, SMOTE, leakage-aware evaluation

Leadership & Achievements

AIUB Research & Development Club **Since January 2025**
Organizer *Dhaka, Bangladesh*

- Helped organize research seminars, methodology workshops, and academic events on machine learning and research methods.
- Co-organized seminars with Prof. Dr. Nowshad Amin (≈150 attendees) and Prof. Pushpa Lata (260+ registrants).

Achievements

- **Winner**, Abstract Presentation Competition, State University
- **1st Runner-Up**, Business Competition
- **Participant**, CUET Three Minute Thesis (3MT), SciBlitz 2.0, CUET